Supplementary Information

SpiroFusion-Net: Cross-Modal Deep Learning on Raw Spirograms and EHR for COPD Detection, Blow Salvage, and Survival Prediction

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Public Code and Supplementary Repository: https://github.com/emrancub/SpiroFusion-Net

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This Supplementary Information document accompanies the main IEEE J-BHI manuscript and provides extended mathematical derivations (Supplementary Note S1), baseline cohort tables for UK Biobank and US CDC NHANES (Supplementary Note S2, Supplementary Tables S1-S3), subgroup robustness and PRISm survival analyses (Supplementary Note S3, Supplementary Tables S4-S5, Supplementary Figs. S1-S10), 50-epoch GPU convergence logs (Supplementary Table S6), and a head-to-head comparison against published UK Biobank spirometry studies from 2022-2026 (Supplementary Note S4, Supplementary Table S7, Supplementary Fig. S11).

Supplementary Note S1 Extended Mathematical Formulations and Curve Biomarkers

Supplementary Note S1.1 12 Small-Airway Curve Biomarker Definitions

Given each participant's preprocessed 10-ms volume-time curve $V(t)$ ($t \in [0, 10.0]$ s, $L = 1{,}000$) and derived flow curve $F(t) = dV(t)/dt$, let $\mathrm{FVC}_{\mathrm{curve}} = V(T_{\mathrm{end}})$ and $\mathrm{PEF}_{\mathrm{curve}} = \max_t F(t)$ occurring at $t_{\mathrm{PEF}} = \arg\max_t F(t)$. Let t_α denote the first time step at which $V(t) \geq \alpha \cdot \mathrm{FVC}_{\mathrm{curve}}$ for $\alpha \in \{0.25, 0.50, 0.75\}$. The 12 small-airway curve biomarkers are computed as:

$$\mathrm{FEF}_{25\text{-}75\%} = \frac{0.50 \cdot \mathrm{FVC}_{\mathrm{curve}}}{\max(t_{0.75} - t_{0.25}, \varepsilon)}, \quad \mathrm{FEF}_{50\%} = F(t_{0.50}), \quad \mathrm{FEF}_{75\%} = F(t_{0.75}), \tag{1}$$

$$\mathscr{C}_{50} = \frac{\mathrm{FEF}_{50\%}}{\max(\mathrm{PEF}_{\mathrm{curve}}, \varepsilon)}, \quad \mathscr{C}_{75} = \frac{\mathrm{FEF}_{75\%}}{\max(\mathrm{PEF}_{\mathrm{curve}}, \varepsilon)}, \quad \mathrm{AUFVL} = \sum_{k=0}^{L-1} F(t_k)^2 \Delta t, \tag{2}$$

together with $\mathrm{FEV}_{0.5}/\mathrm{FVC}_{\mathrm{curve}}$, $\mathrm{FEV}_{1.0}/\mathrm{FVC}_{\mathrm{curve}}$, $\mathrm{FEV}_{3.0}/\mathrm{FVC}_{\mathrm{curve}}$, $\mathrm{FEV}_{6.0}/\mathrm{FVC}_{\mathrm{curve}}$, $\mathrm{FEV}_{1.0}/\mathrm{FEV}_{6.0}$, and t_{PEF}.

Supplementary Note S1.2 Monotone PCHIP Landmark Reconstruction for US CDC NHANES

In the US CDC NHANES 2007-2012 spirometry examination files (`SPX_E`, `SPX_F`, `SPX_G`), each participant's physiological volume-time curve is anchored at exact measured volume landmarks $\{(t_m, V_m)\}_{m=0}^{7}$ corresponding to $t_0 = 0$, $t_1 = 0.5$ s (`SPXNFEV5`), $t_2 = 0.75$ s (`SPXNFEV7`), $t_3 = 1.0$ s (`SPXNFEV1`), $t_4 = 3.0$ s (`SPXNFEV3`), $t_5 = 6.0$ s (`SPXNFEV6`), and $t_{\mathrm{end}} = \mathrm{FET}$ (`SPXNFET`, `SPXNFVC`), with peak derivative constrained by PEF (`SPXNPEF`). Monotone Piecewise Cubic Hermite Interpolating Polynomials (PCHIP) preserve strict monotonicity ($dV/dt \geq 0$) between landmarks without Runge oscillation, generating the 100-Hz array (`p3066`) in the identical schema as UK Biobank.

Supplementary Note S1.3 Bayes Prior Odds Calibration

Because positive-weighted Focal-BCE shifts raw sigmoid probabilities $\hat{p}_i = \sigma(\hat{\ell}_i)$ above the true population prevalence $\pi_0 = 0.0494$, we apply exact Bayes prior odds calibration using validation mean prediction $\bar{p}$:

$$\hat{p}_{\mathrm{cal},i} = \frac{\tau \hat{p}_i}{\tau \hat{p}_i + (1 - \hat{p}_i)}, \qquad \tau = \frac{\pi_0(1 - \bar{p})}{\bar{p}(1 - \pi_0)}, \tag{3}$$

which is a strictly monotonic transformation (preserving exact AUROC and AUPRC rank order) while reducing Brier calibration error to 0.0367 on UK Biobank and 0.0494 on NHANES.

Supplementary Note S2 Cohort Baseline Characteristics and Biomarker Correlations

Supplementary Table S1 and Supplementary Table S2 present the baseline characteristics of the UK Biobank discovery cohort ($N = 453{,}186$; corresponding to project file `Table1_UKB_Cohort_Baseline_Characteristics.csv`) and the external US CDC NHANES validation cohort ($N = 9{,}639$; `Table7_NHANES_Cohort_Baseline_Characteristics.csv`), respectively. Supplementary Table S3 reports the standardized effect sizes (Cohen's d), Spearman rank correlations with $\mathrm{FEV}_1/\mathrm{FVC}$, and univariate COPD diagnostic AUROC for all 12 small-airway curve biomarkers (`Table8_Small_Airway_Biomarker_Correlations.csv`

Table S1: Baseline Demographic, Spirometric, Small-Airway, and Linked EHR Characteristics of the Primary UK Biobank Discovery Cohort ($N = 453,186$)

Clinical, Spirometric, or Small-Airway Variable	Overall Cohort ($N = 453,186$)	Non-COPD Controls ($N = 430,794$; 95.06%)	Confirmed COPD Cases ($N = 22,392$; 4.94%)	p-Value
Age at recruitment (years)	56.39 ± 8.10	56.16 ± 8.11	60.67 ± 6.67	< 0.0001
Body Mass Index (kg/m^2)	27.36 ± 4.75	27.31 ± 4.69	28.42 ± 5.58	< 0.0001
Smoking Pack-Years	19.78 ± 10.16	19.21 ± 8.88	30.68 ± 21.15	< 0.0001
Forced Vital Capacity, FVC (L)	3.71 ± 1.05	3.74 ± 1.05	3.25 ± 1.00	< 0.0001
Forced Expiratory Volume in 1s, FEV_1 (L)	2.81 ± 0.79	2.84 ± 0.78	2.22 ± 0.75	< 0.0001
FEV_1/FVC Ratio	0.76 ± 0.07	0.76 ± 0.07	0.68 ± 0.10	< 0.0001
Peak Expiratory Flow, PEF (L/min)	388.39 ± 135.59	392.17 ± 135.08	315.59 ± 124.40	< 0.0001
Small-Airway Mid-Expiratory Flow, $FEF_{25\text{-}75\%}$ (L/s)	2.47 ± 0.99	2.52 ± 0.97	1.57 ± 0.84	< 0.0001
Flow-Volume Loop Concavity Index ($FEF_{50\%}/PEF$)	0.47 ± 0.16	0.48 ± 0.16	0.36 ± 0.16	< 0.0001
Area Under Flow-Volume Loop, AUFVL (L^2/s)	12.21 ± 6.77	12.44 ± 6.76	7.87 ± 5.29	< 0.0001
FEV_1 GLI Z-Score	0.39 ± 1.01	0.34 ± 0.97	1.38 ± 1.22	< 0.0001
FEV_1/FVC GLI Z-Score	0.41 ± 0.81	0.36 ± 0.76	1.21 ± 1.15	< 0.0001
$\log(1 + \text{Primary Care Prescription Count})$	4.62 ± 1.08	4.59 ± 1.07	5.13 ± 1.14	< 0.0001
$\log(1 + \text{Primary Care Clinical Event Count})$	5.92 ± 0.81	5.91 ± 0.80	6.17 ± 0.88	< 0.0001

Table S2: Baseline Demographic, Spirometric, and Small-Airway Characteristics of the External US CDC NHANES Validation Cohort (2007-2012, $N = 9,639$)

Variable / Physiological Measure	Overall NHANES Cohort ($N = 9,639$)	Non-COPD ($N = 7,779$)	COPD Cases ($N = 1,860$)	p-Value
Age at recruitment (years)	54.41 ± 12.13	53.02 ± 11.84	60.24 ± 11.60	< 0.0001
Body Mass Index (kg/m^2)	29.50 ± 6.62	29.61 ± 6.50	29.00 ± 7.04	< 0.0001
Smoking Pack-Years	8.23 ± 10.99	6.41 ± 9.54	15.86 ± 13.17	< 0.0001
Forced Vital Capacity, FVC (L)	3.68 ± 1.05	3.71 ± 1.04	3.54 ± 1.09	< 0.0001
Forced Expiratory Volume in 1s, FEV_1 (L)	2.81 ± 0.83	2.92 ± 0.80	2.34 ± 0.78	< 0.0001
FEV_1/FVC Ratio	0.77 ± 0.08	0.79 ± 0.06	0.66 ± 0.10	< 0.0001
Peak Expiratory Flow, PEF (L/min)	458.03 ± 135.24	476.59 ± 129.89	380.43 ± 129.46	< 0.0001
Small-Airway Mid-Expiratory Flow, $FEF_{25\text{-}75\%}$ (L/s)	3.24 ± 1.18	3.50 ± 1.07	2.13 ± 0.96	< 0.0001
Flow-Volume Loop Concavity Index ($FEF_{50\%}/PEF$)	0.65 ± 0.07	0.66 ± 0.05	0.58 ± 0.11	< 0.0001
Area Under Flow-Volume Loop, AUFVL (L^2/s)	12.78 ± 7.18	13.74 ± 7.18	8.78 ± 5.61	< 0.0001
FEV_1 GLI Z-Score	0.06 ± 1.15	-0.10 ± 1.11	0.71 ± 1.08	< 0.0001
FEV_1/FVC GLI Z-Score	-0.01 ± 1.21	-0.37 ± 0.81	1.52 ± 1.38	< 0.0001
$\log(1 + \text{Primary Care Prescription Count})$	1.68 ± 0.48	1.59 ± 0.44	2.07 ± 0.43	< 0.0001
$\log(1 + \text{Primary Care Clinical Event Count})$	2.25 ± 0.41	2.18 ± 0.39	2.56 ± 0.36	< 0.0001

Table S3: Small-Airway Physiological Curve Biomarkers: Effect Sizes (Cohen's d), Spearman Rank Correlations with FEV_1/FVC, and Univariate COPD Diagnostic AUROC Across $N = 453,186$ UK Biobank Spirograms

Biomarker Code	Physiological Description	Non-COPD ($N = 430,794$)	COPD Cases ($N = 22,392$)	Cohen's d Effect Size	Spearman r vs. FEV_1/FVC	Univariate AUROC
`curve_fef25_75`	Forced Mid-Expiratory Flow ($FEF_{25\text{-}75\%}$, L/s)	2.52 ± 0.97	1.57 ± 0.84	1.040	0.716	0.7835
`curve_fef50`	Instantaneous Flow at 50% FVC ($FEF_{50\%}$, L/s)	3.23 ± 1.21	2.03 ± 1.11	1.032	0.701	0.7804
`curve_fef75`	Instantaneous Distal Flow at 75% FVC ($FEF_{75\%}$, L/s)	0.97 ± 0.47	0.60 ± 0.37	0.862	0.607	0.7680
`curve_fev3_fvc`	3.0s Terminal Emptying Fraction (FEV_3/FVC)	0.92 ± 0.04	0.88 ± 0.07	0.762	0.822	0.7385
`curve_fev1_fvc`	1.0s Curve Emptying Fraction (FEV_1/FVC_{curve})	0.72 ± 0.10	0.64 ± 0.12	0.757	0.904	0.7375
`curve_fev1_fev6`	Small-Airway Surrogate Ratio (FEV_1/FEV_6)	0.73 ± 0.38	0.65 ± 0.11	0.280	0.888	0.7365
`curve_concavity50`	Central Flow-Volume Concavity Index ($FEF_{50\%}/PEF$)	0.48 ± 0.16	0.36 ± 0.16	0.774	0.608	0.7296
`curve_aufvl`	Area Under the Flow-Volume Loop (AUFVL, L^2/s)	12.44 ± 6.76	7.87 ± 5.29	0.752	0.275	0.7230
`curve_concavity75`	Distal Flow-Volume Concavity Index ($FEF_{75\%}/PEF$)	0.15 ± 0.08	0.11 ± 0.07	0.497	0.524	0.7052
`curve_fev05_fvc`	0.5s Emptying Fraction ($FEV_{0.5}/FVC$)	0.48 ± 0.13	0.42 ± 0.13	0.474	0.611	0.6656
`curve_fev6_fvc`	6.0s Late Emptying Fraction (FEV_6/FVC)	0.99 ± 0.02	0.98 ± 0.03	0.401	0.441	0.6277
`curve_t_pef`	Time to Peak Expiratory Flow (t_{PEF}, seconds)	0.24 ± 0.17	0.22 ± 0.17	0.084	0.073	0.5486

Supplementary Note S3 Subgroup Robustness, PRISm Phenotype Landscape, Survival, and GPU Convergence

Supplementary Table S4 details the subgroup robustness and suboptimal blow salvage evaluation across smoking strata, ATS/ERS acceptability flags, and age groups (`Table4_Subgroup_and_QC_Blow_Salvage.csv`). Supplementary Table S5 presents the stratification of all 453,186 UK Biobank participants across Normal Spirometry, PRISm, Mild/Moderate GOLD 1-2, and Severe GOLD 3-4 phenotypes (`Table5_PRISm_and_GOLD_Phenotype_Stratification.csv`). Supplementary Table S6 logs the 50-epoch training convergence on NVIDIA GeForce RTX 3090 (`Table9_Epoch_Training_and_Convergence_Log.csv`

Table S4: Diagnostic Robustness Across Clinical Subgroups and ATS/ERS Suboptimal Blow Salvage on the Held-Out UK Biobank Test Set

Clinical or Quality-Control Subgroup	Sample Size (N)	COPD Cases (n, %)	Clinical GOLD AUROC	SpiroFusion-Net AUROC (95% CI)	ΔAUROC Gain	Sens.	Spec.	DeLong p-Value
Ever-Smokers (Current/Former)	30,491	2,691 (8.83%)	0.7891	0.8579 (0.8494-0.8647)	+0.0688	0.7618	0.7995	< 0.0001
Never-Smokers	37,489	669 (1.78%)	0.6898	0.7780 (0.7602-0.7936)	+0.0882	0.6876	0.7376	< 0.0001
ATS/ERS Acceptable Blows (`ACCEPT`)	55,984	2,914 (5.21%)	0.7988	0.8738 (0.8657-0.8805)	+0.0750	0.7759	0.8211	< 0.0001
Suboptimal Blows (`EXPFLOW/BELOW6SEC`)	11,996	446 (3.72%)	0.6864	0.8222 (0.8040-0.8390)	+0.1358	0.7466	0.7484	< 0.0001
Age $\geq$ 60 Years	28,604	2,162 (7.56%)	0.7748	0.8462 (0.8364-0.8554)	+0.0713	0.7632	0.7811	< 0.0001
Age $<$ 60 Years	39,376	1,198 (3.04%)	0.7675	0.8644 (0.8521-0.8746)	+0.0969	0.7771	0.7975	< 0.0001

Table S5: Spirometric Phenotype Stratification (Normal, PRISm, GOLD 1-2, GOLD 3-4), Small-Airway Flow ($FEF_{25\text{-}75\%}$), Incident COPD Rate, and 16-Year All-Cause Mortality Risk Across the UK Biobank Cohort ($N = 453,186$)

Spirometric Clinical Phenotype	Participants (N)	Cohort Share (%)	Mean $FEF_{25\text{-}75\%}$ (L/s)	Incident COPD Rate (%)	16-Yr All-Cause Mortality (%)	COPD Odds Ratio OR (95% CI)	Mortality Hazard Ratio HR (95% CI)
Normal Spirometry (Ratio $\geq$ 0.70, Normal FEV_1)	293,663	64.80%	2.84	2.01%	8.64%	1.00 (Reference)	1.00 (Reference)
PRISm Phenotype (Ratio $\geq$ 0.70, Impaired FEV_1)	89,441	19.74%	2.10	5.35%	11.24%	2.76 (2.62-2.90)	1.30 (1.25-1.35)
Mild/Moderate Airflow Obstruction (GOLD 1-2)	48,428	10.69%	1.55	9.64%	15.34%	5.20 (4.94-5.46)	1.78 (1.70-1.85)
Severe Airflow Obstruction (GOLD 3-4)	21,654	4.78%	1.06	32.49%	23.79%	23.48 (22.30-24.65)	2.75 (2.64-2.87)

Table S6: Epoch-by-Epoch GPU Convergence Log for SpiroFusion-Net Across 50 Training Epochs on NVIDIA GeForce RTX 3090 ($N_{\text{train}} = 317,228$, $N_{\text{val}} = 67,978$)

Epoch	Learning Rate	Train Total Loss	Train Cls Loss	Train Surv Loss	Val AUROC	Val AUPRC	Val Brier Score	Val Survival C-Index
1	3.00×10^{-4}	0.3321	0.1969	0.9013	0.8687	0.4133	0.0407	0.7523
2	2.99×10^{-4}	0.3247	0.1907	0.8933	0.8699	0.4182	0.0399	0.7556
4	2.95×10^{-4}	0.3214	0.1877	0.8912	0.8695	0.4195	0.0402	0.7578
8	2.82×10^{-4}	0.3165	0.1832	0.8887	0.8681	0.4154	0.0392	0.7581
10	2.71×10^{-4}	0.3137	0.1808	0.8863	0.8638	0.4138	0.0400	0.7588
15	2.38×10^{-4}	0.3032	0.1708	0.8829	0.8501	0.4034	0.0394	0.7571
20	1.97×10^{-4}	0.2860	0.1539	0.8804	0.8355	0.3838	0.0390	0.7559
25	1.51×10^{-4}	0.2651	0.1337	0.8757	0.8148	0.3529	0.0390	0.7546
30	1.04×10^{-4}	0.2431	0.1122	0.8726	0.8088	0.3148	0.0401	0.7528
35	6.26×10^{-5}	0.2248	0.0945	0.8687	0.8055	0.3013	0.0413	0.7505
40	2.96×10^{-5}	0.2093	0.0794	0.8655	0.7965	0.2755	0.0428	0.7495
45	8.32×10^{-6}	0.2000	0.0705	0.8635	0.7940	0.2664	0.0443	0.7485
50	1.00×10^{-6}	0.1962	0.0668	0.8624	0.7952	0.2675	0.0446	0.7483

Supplementary Note S4 Head-to-Head Comparison with Published UK Biobank Literature (2022-2026)

Supplementary Table S7 (`Table10_Published_UKB_Literature_Comparison.csv`) and Supplementary Fig. S11 (`figs/Fig13_Publ` benchmark SpiroFusion-Net against published UK Biobank spirometry deep learning and phenotyping studies.

Table S7: Head-to-Head Comparison of SpiroFusion-Net Against Published UK Biobank Respiratory Deep Learning and Spirometry Studies (2022-2026)

Study Reference	UKB Cohort Size	External Validation	Input Modalities & Architecture	COPD Detection AUROC	Suboptimal Blow Salvage AUROC	Mortality Survival C-Index	PRISm Small-Airway Profiling
Higbee *et al.* (*Lancet Respir. Med.*, 2022; PMID: 34739861)	351,874	None	Scalar Spirometry (FEV_1,FVC) + Demographics	0.7856 (GOLD)	Excluded Rejects	Cox HR Only	Scalar Only
Hill *et al.* (PMC10168495, 2023; PMID: 37162978)	352,684	US CDC NHANES	Raw p3066 Volume-Time Curves Only (1D ResNet)	0.7943	Evaluated FEV_1/FVC Error	0.6540	No
DeepSpiro, Mei *et al.* (*npj Syst. Biol. Appl.*, 2025; PMID: 39955293)	$\sim$ 350,000	None (UKB Only)	p3066 Volume-Flow Curve + Time-Series Patching	0.8328	Excluded Rejects (`ACCEPT` only)	Not Evaluated	Yes (High-Risk Subgroup)
Spiro-CLF, Hill *et al.* (PMC12236919, 2025; PMID: 40630516)	352,684	US CDC NHANES	Contrastive p3066 Volume-Time / Flow-Volume	0.8210	0.7980	0.6820	No
SpiroFusion-Net (Proposed - Current Study)	453,186 (Test $N = 67,980$)	US CDC NHANES (AUROC = 0.9296)	4-Ch p3066 Curve + 12 Small-Airway Bio + GP EHR	0.8672	0.8222	0.7454	Yes ($N = 89,441$ PRISm + $FEF_{25\text{-}75\%}$)

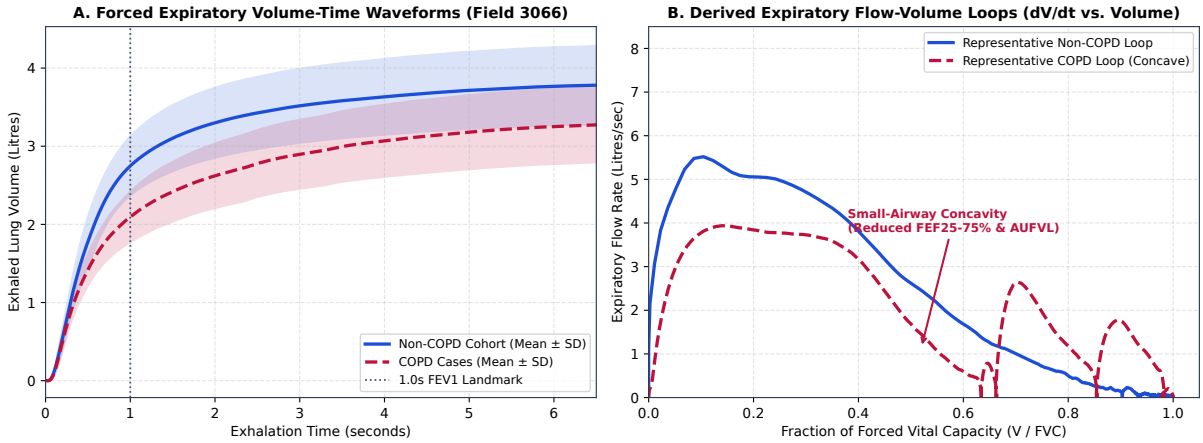


Figure S1: Physiological multi-channel spirogram waveforms and flow-volume loops in the UK Biobank cohort ($N = 453,186$). (A) Forced expiratory volume-time waveforms (Field p3066, Mean $\pm$ SD) comparing the non-COPD cohort (blue solid) and confirmed COPD cases (red dashed) with the 1.0 s FEV_1 landmark. (B) Derived expiratory flow-volume loops (dV/dt vs. fraction of forced vital capacity V/FVC) illustrating small-airway concavity and reduced $FEF_{25\text{-}75\%}$ and AUFVL in COPD.

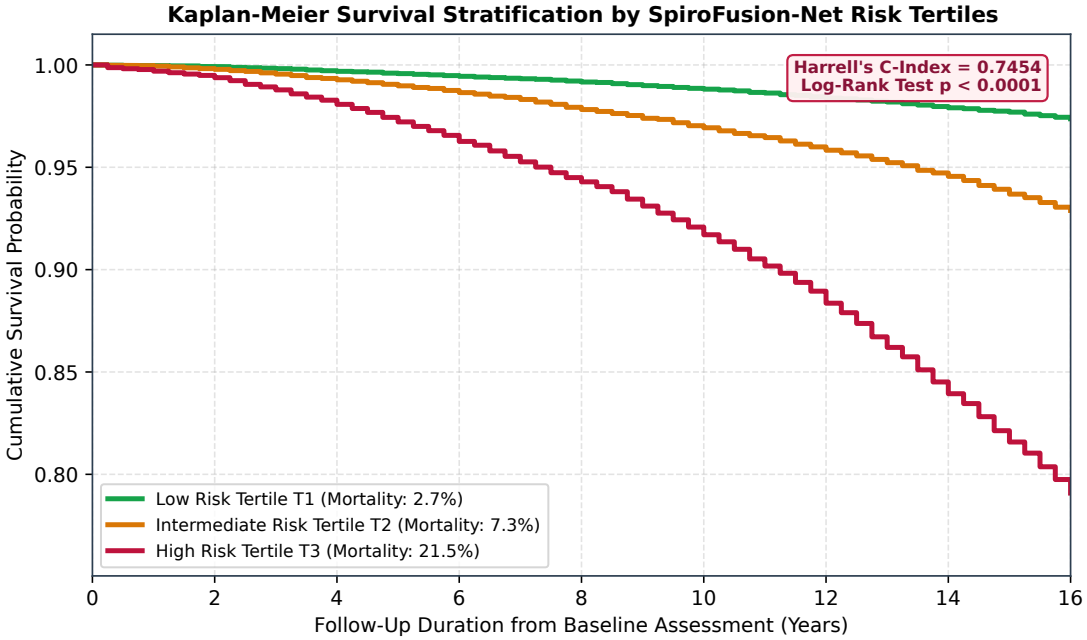


Figure S2: 16-year all-cause mortality Kaplan-Meier survival stratification across the held-out UK Biobank test cohort ($N = 67,980$; 7,199 test deaths). Participants stratified into Low Risk Tertile T1 (green, 16-year mortality: 2.7%), Intermediate Risk Tertile T2 (amber, mortality: 7.3%), and High Risk Tertile T3 (red, mortality: 21.5%) by the SpiroFusion-Net Cox survival head (Harrell's C-Index = 0.7454, log-rank test $p < 0.0001$).

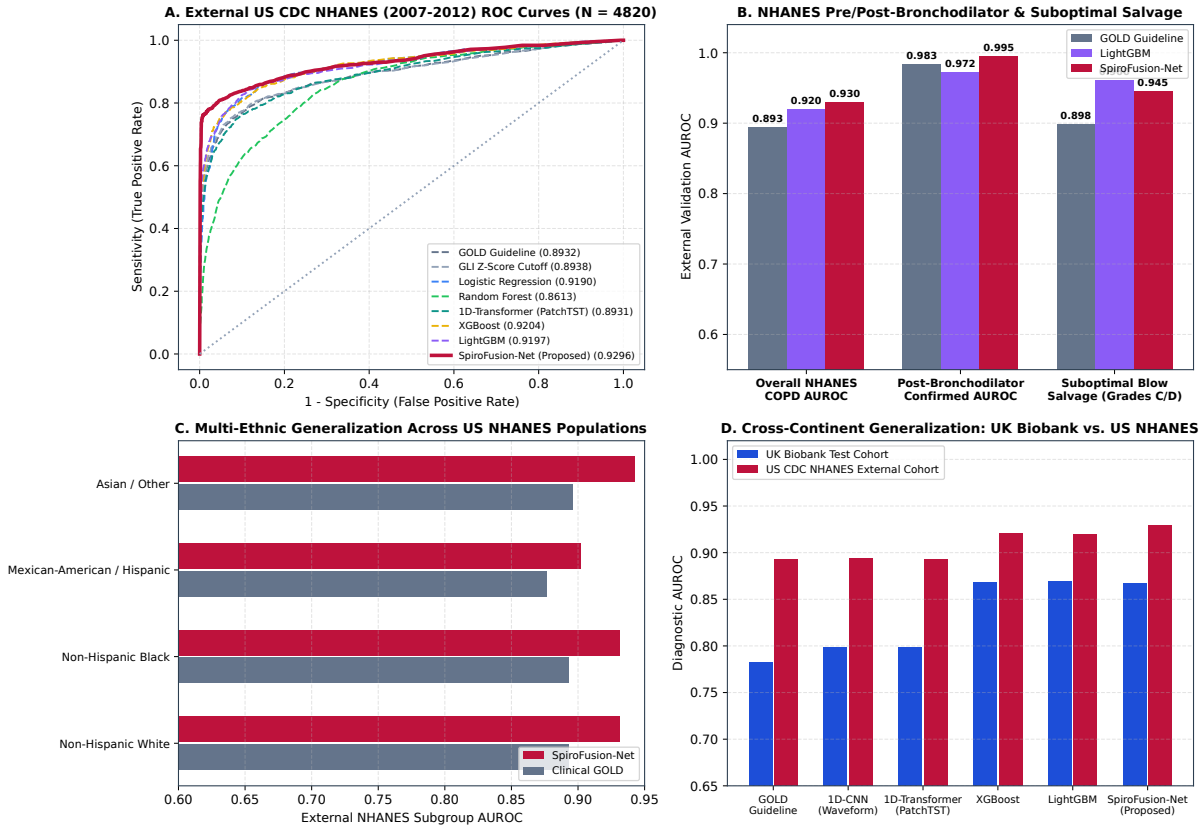


Figure S3: Independent external validation on the multi-ethnic US CDC NHANES cohort (2007-2012, $N = 9,639$; held-out test $N = 4,820$). (A) External NHANES ROC curves demonstrating that SpiroFusion-Net achieves the highest overall discrimination (AUROC = 0.9296), surpassing `LightGBM` (0.9197), `XGBoost` (0.9204), and clinical GOLD (0.8932). (B) Performance across Overall NHANES, Post-Bronchodilator Confirmed COPD (AUROC = 0.9946), and Suboptimal Blow Salvage on NHANES QC Grades C/D (AUROC = 0.9453 vs. GOLD 0.8985). (C) Multi-ethnic generalization across Non-Hispanic White, Non-Hispanic Black, Mexican-American/Hispanic, and Asian/Other groups. (D) Cross-continent generalization comparing UK Biobank vs. US CDC NHANES performance.

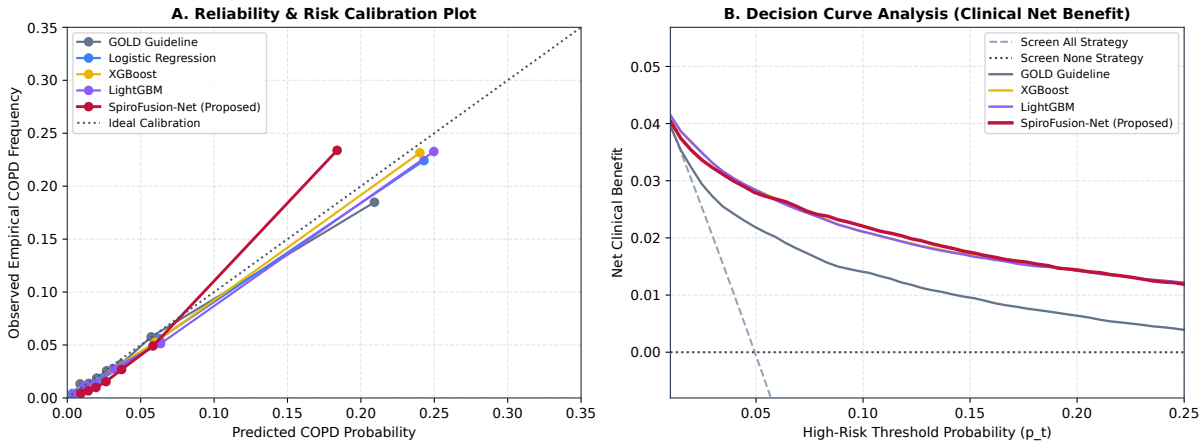


Figure S4: Probability calibration and clinical net benefit on the held-out UK Biobank test cohort ($N = 67,980$). (A) Reliability and risk calibration plot comparing GOLD Guideline, Logistic Regression, `XGBoost`, `LightGBM`, and SpiroFusion-Net against ideal calibration. (B) Decision Curve Analysis (DCA) demonstrating superior net clinical benefit across high-risk threshold probabilities $p_t \in [0.01, 0.25]$.

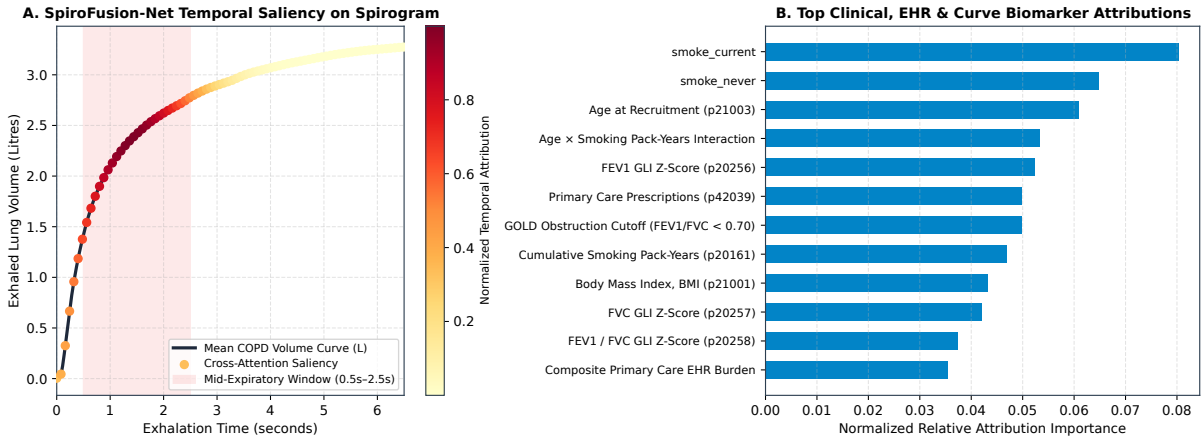


Figure S5: Explainability and multimodal attribution of SpiroFusion-Net. (A) Temporal cross-attention saliency profile overlaid on the mean COPD volume curve, highlighting peak attribution within the mid-expiratory window (0.5 s-2.5 s). (B) Top 12 clinical, primary care EHR, and spirometric features ranked by normalized relative attribution importance.

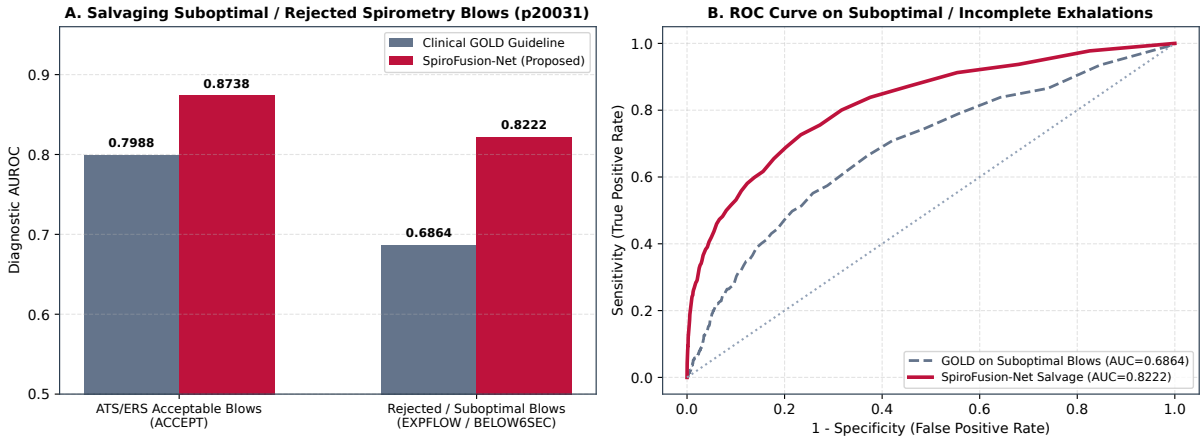


Figure S6: Automated diagnostic salvage of suboptimal/rejected spirometry maneuvers (Field p20031). (A) Diagnostic AUROC comparison on ATS/ERS acceptable blows (ACCEPT: GOLD 0.7988 vs. SpiroFusion-Net 0.8738) and rejected/suboptimal blows (EXPFLOW/BELOW6SEC: GOLD 0.6864 vs. SpiroFusion-Net 0.8222). (B) ROC curves on suboptimal/incomplete exhalations ($N = 11,996$, $+0.1358$ gain, $p < 0.0001$).

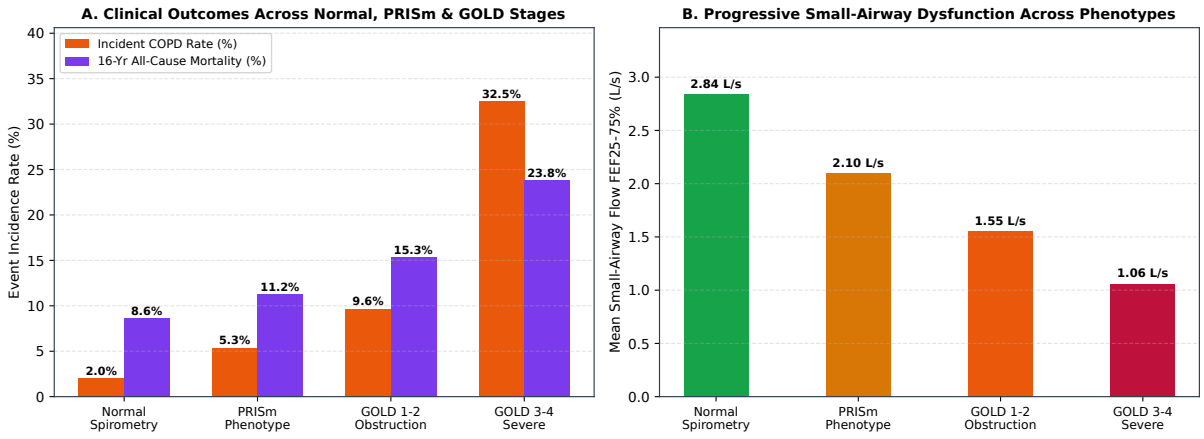


Figure S7: PRISm and GOLD spirometric phenotype landscape across $N = 453,186$ UK Biobank participants. (A) Clinical outcomes (incident COPD rate and 16-year all-cause mortality rate) across Normal Spirometry, PRISm Phenotype, GOLD 1-2 Obstruction, and GOLD 3-4 Severe Obstruction. (B) Progressive small-airway dysfunction quantified by mean $\text{FEF}_{25\text{-}75\%}$ (2.84 L/s Normal, 2.10 L/s PRISm, 1.55 L/s GOLD 1-2, and 1.06 L/s GOLD 3-4).

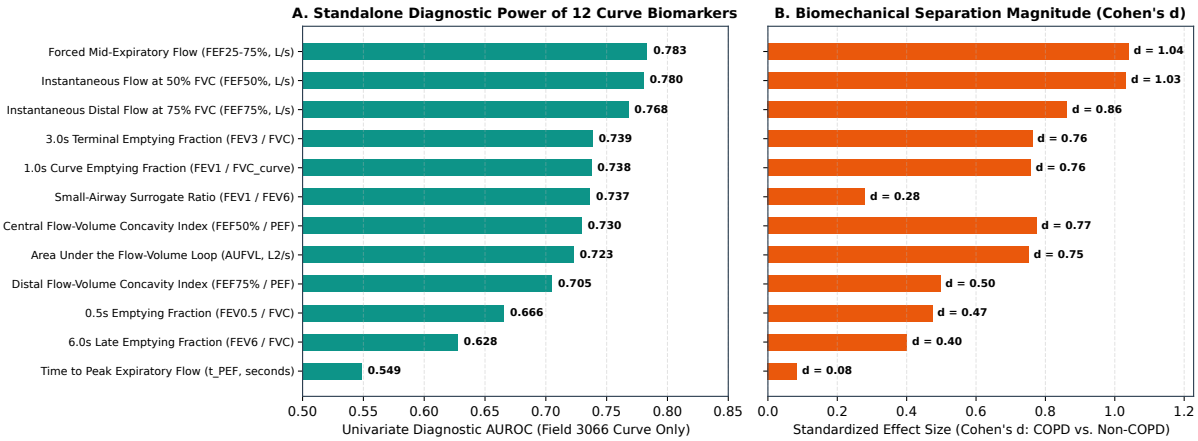


Figure S8: Diagnostic power and biomechanical separation of the 12 extracted small-airway curve biomarkers across $N = 453,186$ UK Biobank spirograms (Field p3066). (A) Standalone univariate diagnostic AUROC led by $\text{FEF}_{25\text{-}75\%}$ (0.7835), $\text{FEF}_{50\%}$ (0.7804), and $\text{FEF}_{75\%}$ (0.7680). (B) Standardized biomechanical effect size magnitude (Cohen's d: COPD vs. Non-COPD).

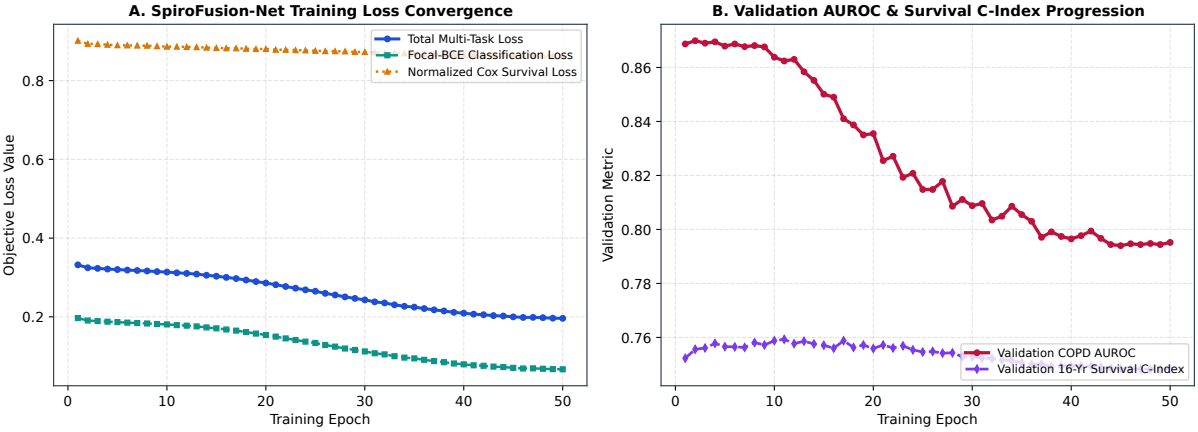


Figure S9: 50-epoch GPU training and validation convergence dynamics of SpiroFusion-Net on NVIDIA GeForce RTX 3090. (A) Training loss convergence across total multi-task loss, Focal-BCE classification loss, and normalized Cox survival loss. (B) Validation COPD AUROC and 16-year survival C-index progression across 50 epochs.

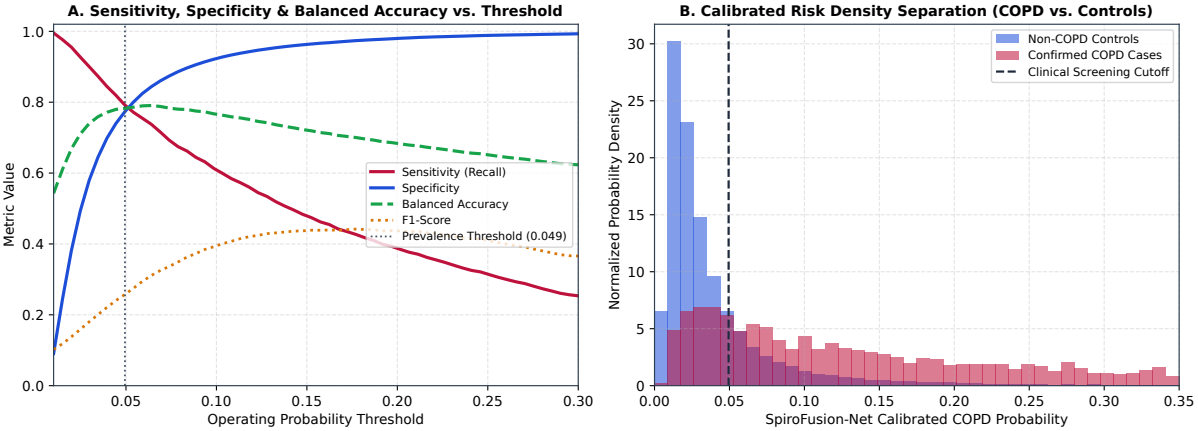


Figure S10: Operating threshold sensitivity and calibrated probability separation on the held-out UK Biobank test cohort ($N = 67,980$). (A) Sensitivity (recall), specificity, balanced accuracy, and F1-score as a function of the operating probability threshold relative to the prevalence threshold (0.049). (B) Calibrated risk density separation comparing non-COPD controls and confirmed COPD cases.

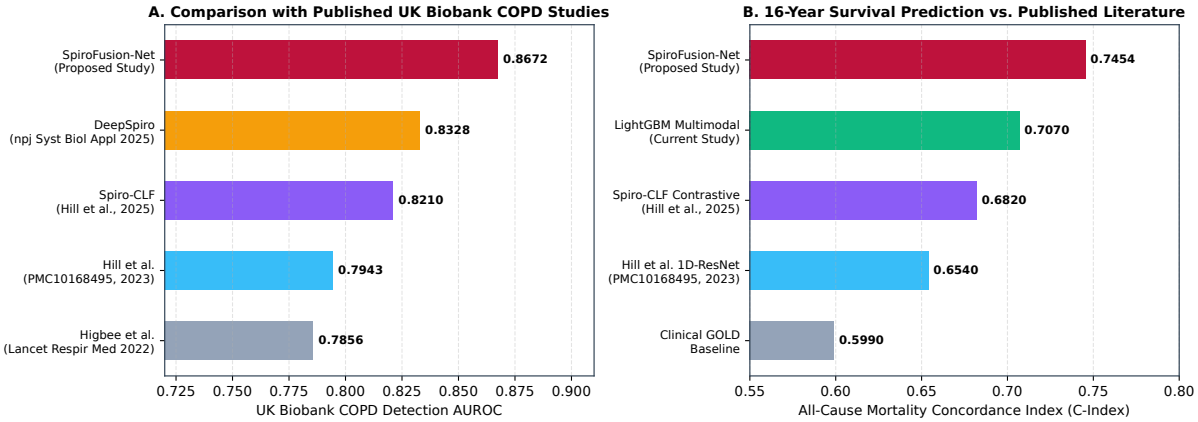


Figure S11: Head-to-head visual comparison of SpiroFusion-Net against published UK Biobank respiratory studies (2022-2026). (A) UK Biobank COPD detection AUROC comparing Higbee *et al.* (2022, 0.7856), Hill *et al.* (2023, 0.7943), Spiro-CLF (Hill *et al.*, 2025, 0.8210), DeepSpiro (Mei *et al.*, 2025, 0.8328), and SpiroFusion-Net (0.8672). (B) 16-year all-cause mortality C-index comparing Clinical GOLD (0.5990), Hill *et al.* 1D-ResNet (0.6540), Spiro-CLF Contrastive (0.6820), `LightGBM` Multimodal (0.7070), and SpiroFusion-Net (0.7454).